

Fundamentals Physics

Twelfth Edition

Halliday

Chapter 5

Force and Motion—I

5.1 Newton's First and Second Laws

Newton's First and Second Laws

Learning Learning Objectives #1 - #4

- Identify that a force is a vector quantity and thus has both magnitude and direction and also components.
- Given two or more forces acting on the same particle, the summation of forces as vectors will yield the net force.
- Identify Newton's first and second laws of motion.
- Identify inertial reference frames.

Newton's First and Second Laws

Learning Learning Objectives #5 - #7

- Sketch a free-body diagram for an object, showing the object as a particle and drawing the forces acting on it as vectors anchored to the particle.
- Apply the relationship between net force on an object, its mass, and the produced acceleration.
- Identify that only external forces on an object can cause the object to accelerate.

Force and Newtonian Mechanics

- **A force:**
 - Is a “push or pull” acting on an object
 - Causes acceleration
- We will focus on Newton's three laws of motion:
 - **Newtonian mechanics** is valid for everyday situations
 - It is not valid for speeds which are an appreciable fraction of the speed of light
 - It is not valid for objects on the scale of atomic structure
 - Can be considered a limiting case in regards to general relativity and quantum mechanics.

Newton's Laws of Motion without Friction

- Before Newtonian mechanics:
 - Some influence (force) was thought necessary to keep a body moving
 - The “natural state” of objects was at rest
- This seems intuitively reasonable (due to friction)
- But envision a **frictionless surface**
 - Does not slow an object
 - The object would keep moving forever at a constant speed
 - Friction is a force!

Newton's First Law: force vs. acceleration

If no force acts on a body, the body's velocity cannot change: that is, the body cannot accelerate.

Characteristics of Forces

- Characteristics of forces:
 - Unit: N, the newton; $1 \text{ N} = 1 \text{ kg m/s}^2$
 - Acceleration of a mass is proportional to the exerted force
 - Forces are vectors
- **Net force** is the vector sum of all forces on an object
- **Principle of superposition for forces:**
 - A net force has the same impact as a single force with identical magnitude and direction
 - So we can restate more correctly:

Net force vs. Acceleration

If no net force acts on a body $\vec{F}_{net} = 0$

the body's velocity cannot change; that is, the body cannot accelerate.

Inertial Reference Frames

- Newton's first law is not true in all frames
- **Inertial frames:**

An inertial reference frame is one in which Newton's laws hold.

(a) a frictionless puck, pushed from the north pole, viewed from space

(b) the same situation, viewed from the ground

Over **long distances** (such as from space), the ground is a **noninertial frame**, but for most practical problems the earth's surface can be considered an inertial reference frame.

In scenario (b), a fictitious force would be needed to explain deflection

Inertial vs. Non-Inertial Reference Frame Graphical Example

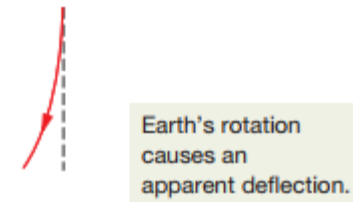
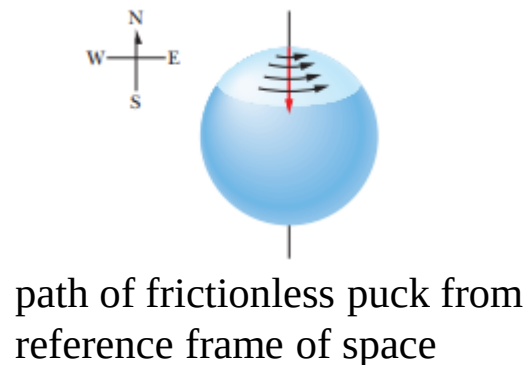


Figure 5.1.2

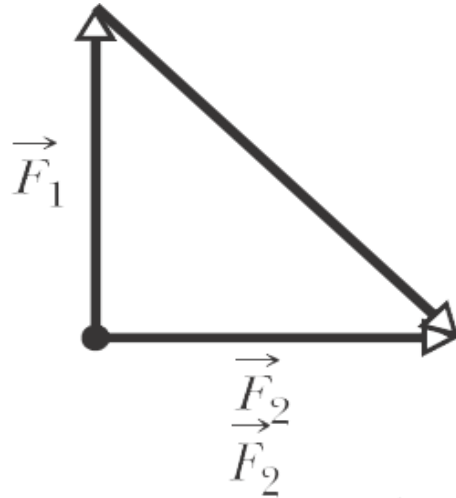
Checkpoint #1 Question Statement

Checkpoint 1

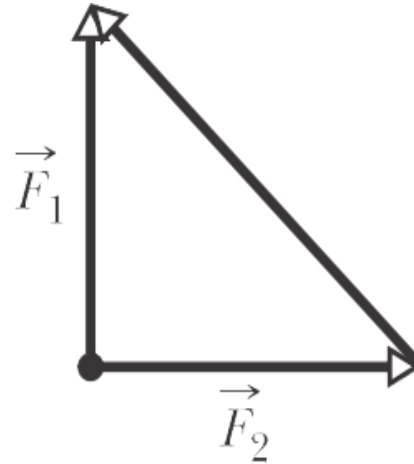
Which of the figure's six arrangements correctly show the vector addition of forces $\vec{F}_1$ and $\vec{F}_2$ to yield the third vector, which is meant to represent their net force $\vec{F}_{net}$

Checkpoint #1 Possible Choices

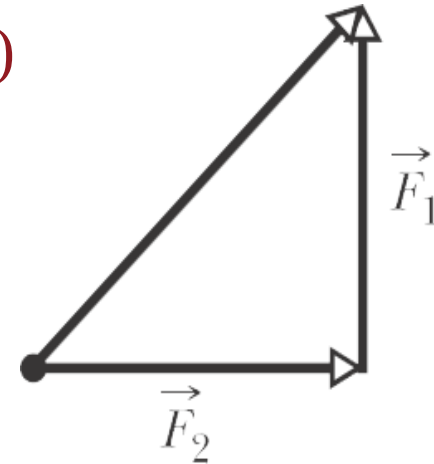
a)



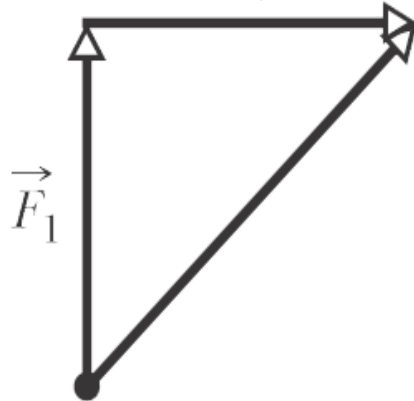
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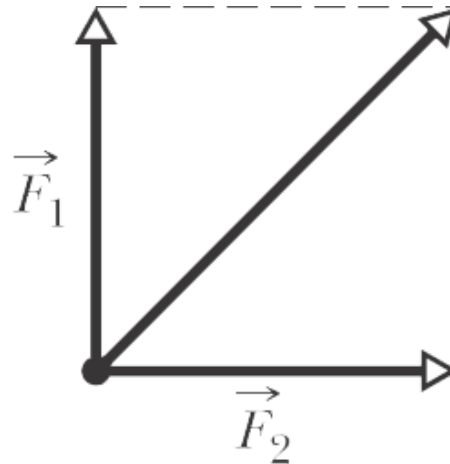
c)



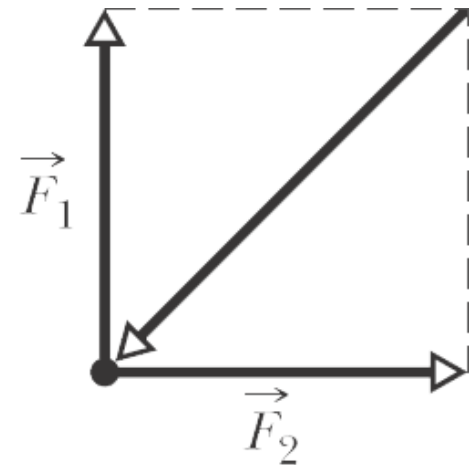
d)



e)



f)



Answer: (c), (d), (e)

Definition of Mass

- What is mass?
 - “the mass of a body is the characteristic that relates a force on the body to the resulting acceleration”
 - Mass is a measure of a body’s resistance to a change in motion (change in velocity)
 - It is not the same as weight, density, size etc.
 - For a given net force, mass is inversely proportional to acceleration.
 - Mass specifies how much inertia an object has.

Effect of Mass on Acceleration

Example Apply an 8.0 N force to various bodies:

- Mass : 1kg $\rightarrow$ acceleration: 8 m/s^2
- Mass : 2kg $\rightarrow$ acceleration: 4 m/s^2
- Mass : 0.5kg $\rightarrow$ acceleration: 16 m/s^2
- Acceleration : $2 \text{ m/s}^2 \rightarrow$ mass: 4kg

Newton's Second Law Statement

- Summarize these behaviors as:

Newton's Second Law: The net force on a body is equal to the product of the body's mass and its acceleration.

- As an equation, we write:

$$\vec{F}_{net} = ma \quad \text{Equation (5.1.1)}$$

- Identify the body in question, and only include forces that act on that body!
- Separate the problem axes (they are independent):

$$F_{net,x} = ma_x, \quad F_{net,y} = ma_y, \quad \text{and} \quad F_{net,z} = ma_z. \quad \text{Equation (5.1.2)}$$

Vector Nature of Force and Acceleration

The acceleration component along a given axis is caused only by the sum of the force components along that same axis, and not by force components along any other axis.

Net Force vs. Body Acceleration

- If the net force on a body is zero:
 - Its acceleration is zero
 - The forces and the body are in equilibrium
 - There may still be forces that are present, but they cancel out!

Units of Force in Newton's Second Law

Table 5.1.1

System	Force	Mass	Acceleration
SI	newton (N)	kilogram (kg)	m/s ²
CGS ^a	dyne	gram (g)	cm/s ²
British ^b	pound (lb)	slug	ft/s ²

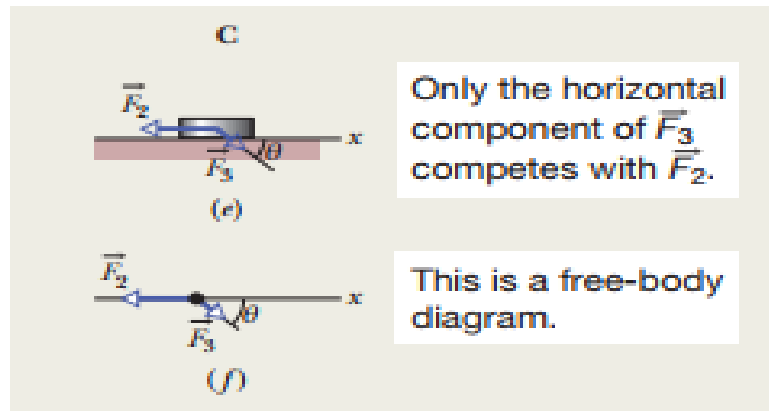
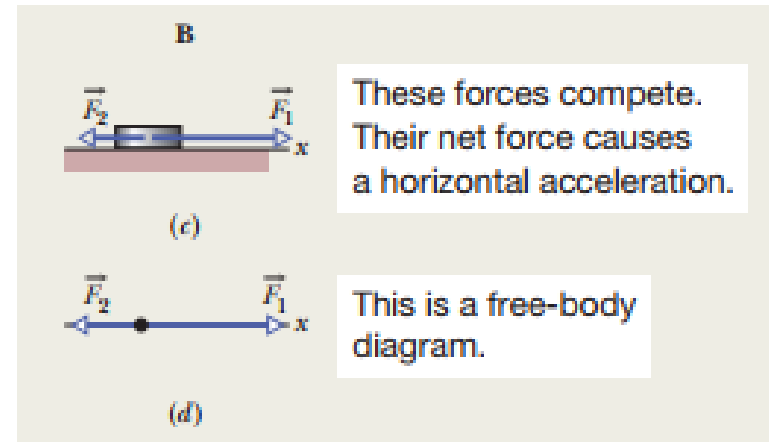
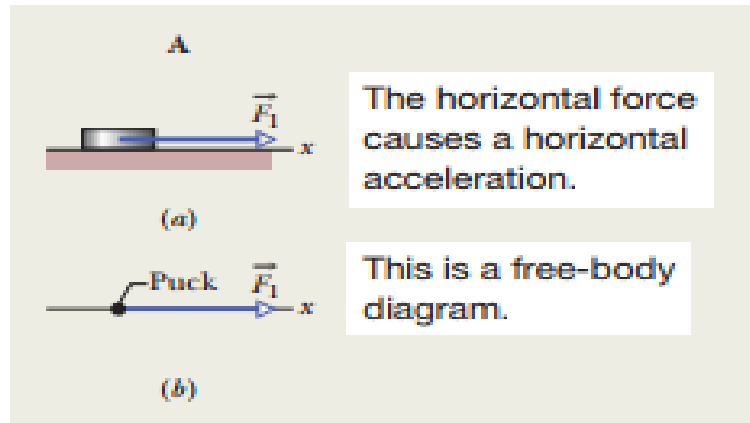
^a1dyne = 1g · cm/s².

^b1lb = 1slug · ft/s².

Introduction to Free Body Diagrams

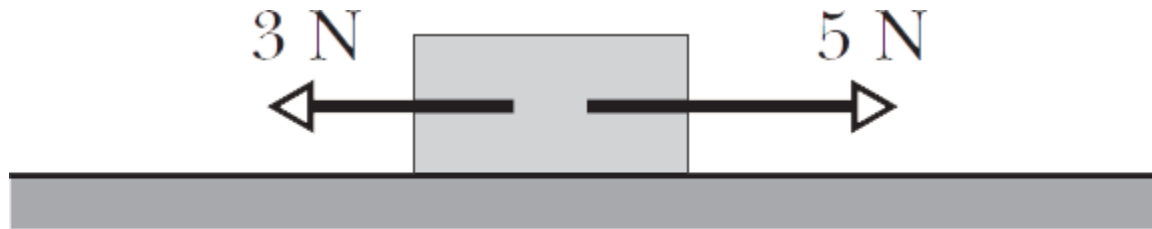
- To solve problems with forces, we often draw a **free body diagram**
- The only body shown is the one we are solving for
- Forces are drawn as vector arrows with their tails on the body
- Coordinate system shown
- Acceleration is Never part of a free body diagram – only forces on a body are present.

Three Examples of Free Body Diagrams



For scenario's **A**, **B**, and **C** the objects and forces shown in the schematics (top drawings) can be represented by free-body diagrams (bottom drawings).

Checkpoint #2 Question



The figure here shows two horizontal forces acting on a block on a frictionless floor. If a third horizontal force $\vec{F}_3$ also acts on the block, what are the magnitude and direction of $\vec{F}_3$ when the block is (a) stationary and (b) moving to the left with a constant speed of 5 m/s?

Answer: $F_3 = 2 \text{ N}$ to the left in both cases

Internal vs. External Forces on Bodies

- A **system** consists of one or more bodies
- Any force on the bodies inside a system exerted by bodies outside the system is an **external force**
- Net force on a system = sum of external forces
- Forces between bodies in a system: **internal forces**
 - Not included in a FBD of the system since internal forces cannot accelerate the system

Note: do not confuse a free body diagram of an entire system with free body diagrams of individual bodies within a system.

Section 5.2 Some Particular Forces

Particular Forces Learning Objectives

#1-#3

1. Determine the magnitude and direction of gravitational force on a mass, for a given free-fall acceleration.
2. Identify that weight is the magnitude of the net force required to prevent a body from falling freely, measured by the frame of ground.
3. Identify that a scale gives an object's weight when the measurement is done in an inertial frame but not in an accelerating frame, where it gives an apparent weight.

Particular Forces Learning Objectives

#4-#6

4. Determine the magnitude and direction of the normal force on an object when the object is pressed or pulled onto a surface.
5. Identify that the force parallel to the surface is a frictional force that appears when the object slides or attempts to slide along the surface.
6. Identify that a tension force is said to pull at both ends of a cord (or a cord-like object) when the cord is taut.

Gravitational Force

- The **gravitational force**:
 - A pull that acts on a body, directed toward a second body
 - Generally we consider situations where the second body is the Earth
- In free fall (y direction, with no drag from the air):

$$-F_g = m(-g) \quad \text{Equation (5.2.1)}$$

$$F_g = mg.$$

- This force still acts on a body at rest!
- We can write it as a vector:

$$\vec{F}_g = -F_g \hat{j} = -mg\hat{j} = m \vec{g} \quad \text{Equation (5.2.2)}$$

Definition of Weight

- The name given to the gravitational force that one body (like the Earth) exerts on an object
 - It is a force measured in newtons (N)
 - It is directed towards the center of mass of a body, which is downwards for an object on the earth's surface.

$$W = F_g \text{ (weight, with ground as inertial frame).}$$

The weight W of a body is equal to the magnitude F_g of the gravitational force on the body.

Weight in Newton's Second Law

Example: To relate weight to mass, consider an apple in free fall. The only force on the apple is the gravitational force which results in an acceleration of g . Applying Newton's 2nd Law

$$F_{\text{net}} = ma \quad \text{where } F_{\text{net}} = F_g = W \quad \text{and } a = g$$

$$F_g = W = mg$$

Thus,

$$W = mg \quad (\text{mass – weight relationship})$$

Measuring Weight

- Measuring weight:
 - Use a balance to compare a body to known masses, find its mass, and compute its weight
 - Use a spring scale that measures weight on a calibrated scale
 - Weight is not the same as mass: a pan balance will read the same for different values of g , a scale will read differently for different values of g
- Weight must be measured when the body is not accelerating vertically
 - E.g., in your bathroom, or on a train
 - But not in an elevator

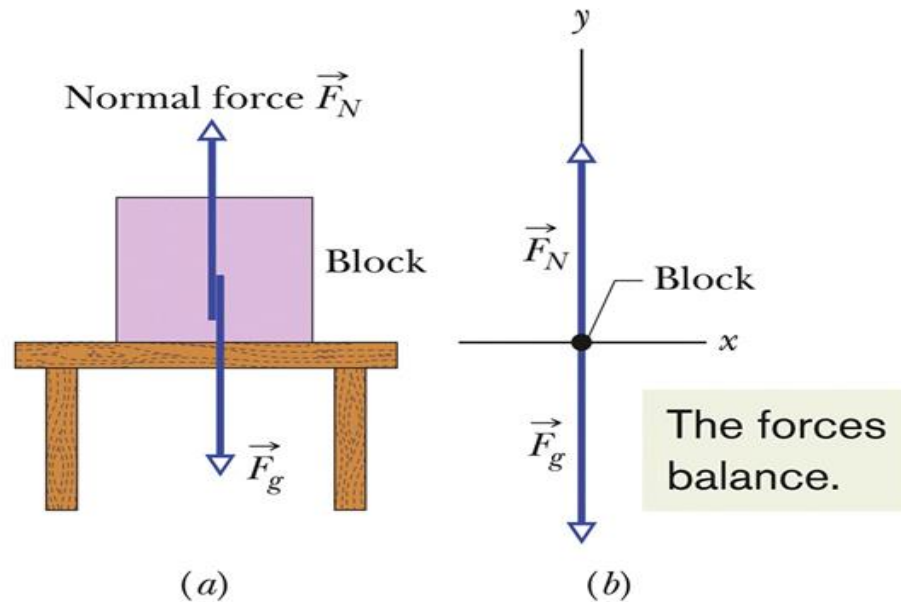
Definition of Normal Force

- The **normal force**:
 - If you are standing on a surface, the push back on you from the surface (due to deformation) is the normal force
 - Normal means perpendicular
- When a body presses against a surface, the surface deforms and pushes on the body with a normal force $\vec{F}_N$ that is perpendicular to the surface. For rigid bodies, deformation is negligible.

Normal Force in a Free Body Diagram

The normal force is the force on the block from the supporting table.

The gravitational force on the block is due to Earth's downward pull.



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Figure 5.2.3

Calculating Normal Force from Statement of Vertical Equilibrium

Example: Normal force for a block resting on a horizontal surface that is:

- Accelerating vertically at a_y :

$$F_N = mg + ma_y = m(g + a_y) \quad \text{Equation (5.2.6)}$$

- Vertically at rest:

$$F_N = mg. \quad \text{Equation (5.2.7)}$$

note that gravitational acceleration g is, by definition, in the y -direction (vertical).

Checkpoint #3

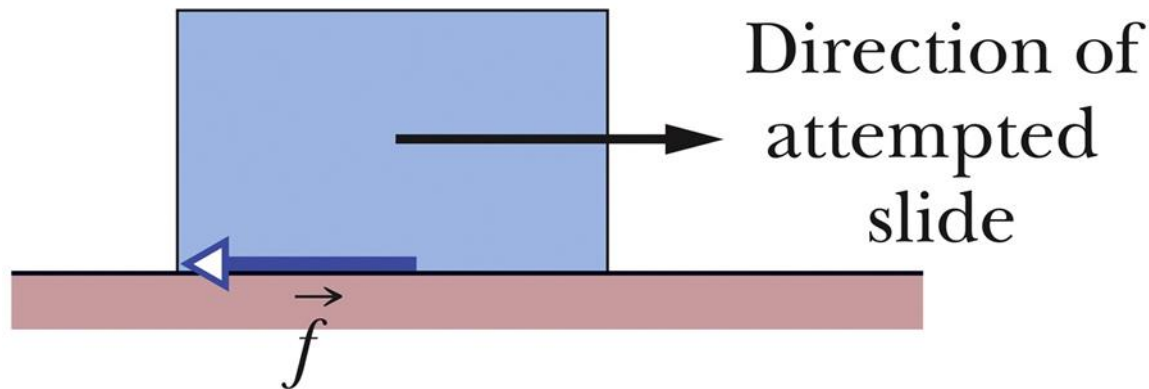
In Fig. 5.2.3, is the magnitude of the normal force $\vec{F}_N$ greater than, less than, or equal to mg if the block and table are in an elevator moving upward (a) at constant speed and (b) at increasing speed?

Answer:

- (a) equal to mg (no acceleration)
- (b) greater than mg (see 5-13, with positive acceleration)

Friction - Key Characteristics

- **Frictional force or friction:**
 - Occurs when one object slides or attempts to slide over another
 - Directed along the surface, opposite to the direction of intended motion



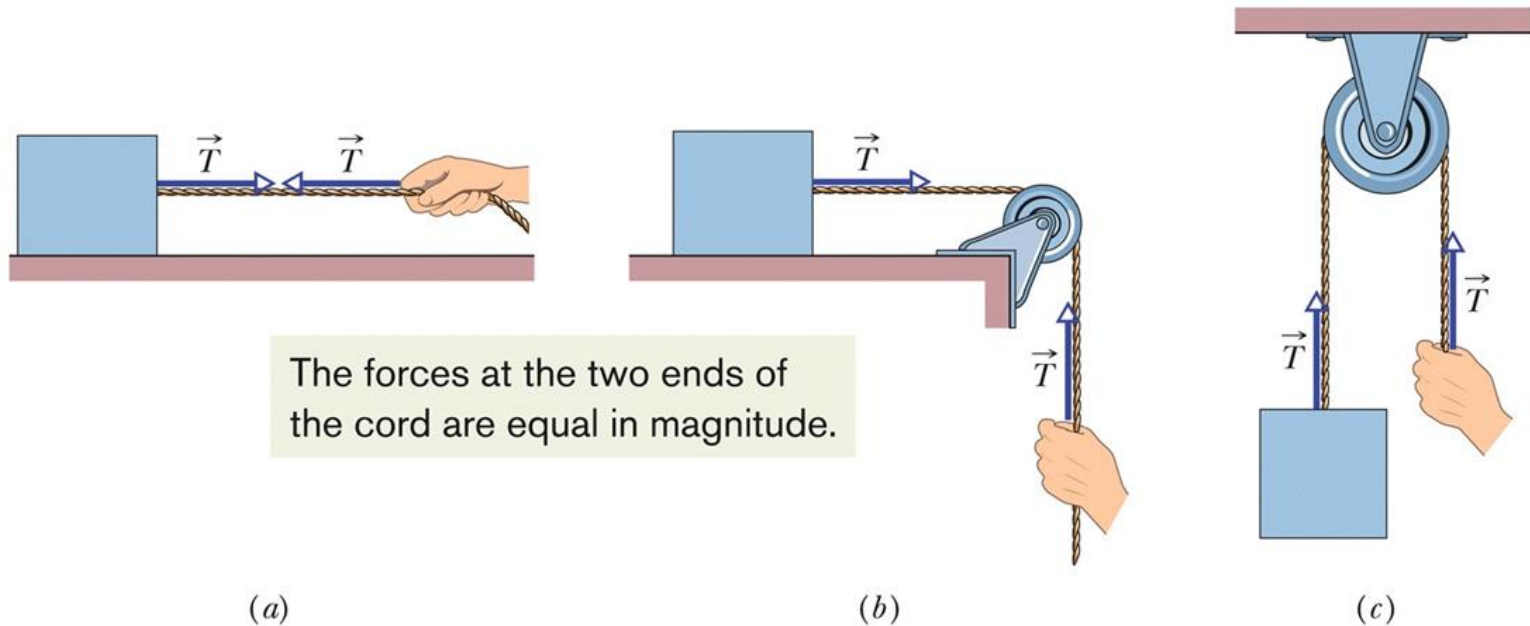
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Figure 5.2.4

Tension - Key Characteristics

- **Tension force:**
 - A cord (or rope, etc.) is attached to a body and pulled taut
 - Cord pulls on the body with force T directed along the cord
 - The cord is said to be under tension
 - The tension in the cord is T
- A massless and unstretchable cord exists only as a connection between two bodies
 - It pulls on both with the same force, T
 - True even if the bodies and cord are accelerating, and even if the cord runs around a massless, frictionless pulley
 - These are useful simplifying assumptions

Tension in Free Body Diagrams



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Figure 5.2.5

Checkpoint #4

The suspended body in Fig. 5.2.5 (c) weighs 75 N. Is T equal to, greater than, or less than 75 N when the body is moving upward (a) at constant speed, (b) at increasing speed, and (c) at decreasing speed?

Answer:

(a) equal to 75 N

(b) greater than 75 N

(c) less than 75 N

Section 5.3 Applying Newton's Laws

Applying Newton's Laws Learning Objectives #1 - #3

1. Identify Newton's third law of motion and third-law of force pairs.
2. For an object that moves vertically or on a horizontal or inclined plane, apply Newton's second law to a free-body diagram of the object.
3. For an arrangement where a system of several objects moves rigidly together, draw a free-body diagram and apply Newton's second law for the individual objects and also for the system taken as a composite object.

Newton's Third Law: Action-Reaction

- Objects interact when they push or pull on each other:

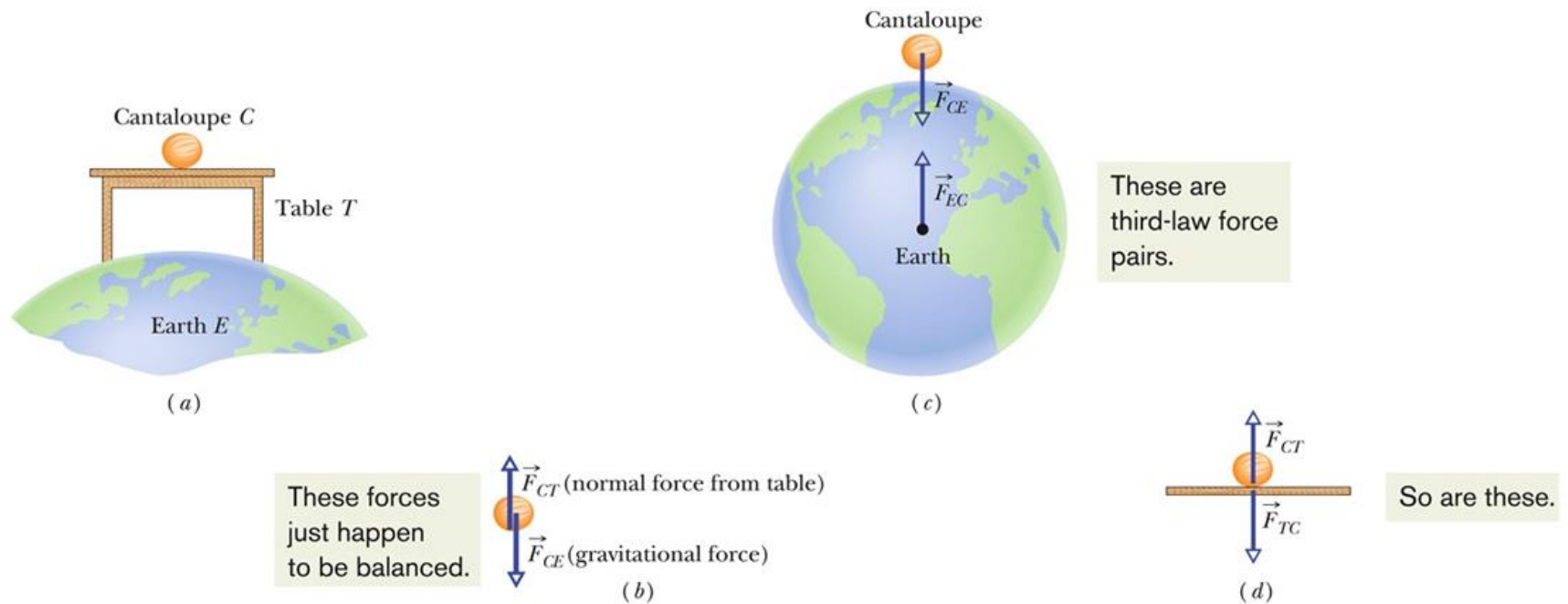
Newton's Third Law: When two bodies interact, the forces on the bodies from each other are always equal in magnitude and opposite in direction.

- We can write this law as a scalar or vector relation:

$$F_{BC} = F_{CB} \quad \vec{F}_{BC} = -\vec{F}_{CB} \quad \text{Equation (5.3.1)}$$

- We call these two forces a **third-law force pair**
- Any time any two objects interact, there is a third-law force pair

Examples of Newton's Third Law



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Figure 5.3.2

Mathematical Expressions from Newton's Third Law

- Third-law force pairs:

$$\vec{F}_{CT} = -\vec{F}_{TC}$$

cantaloupe - table interaction

$$\vec{F}_{CE} = -\vec{F}_{EC}$$

cantaloupe - earth interaction

- This includes the gravitational forces between Earth and the cantaloupe!

Checkpoint #5

Suppose that the cantaloupe and table of Fig. 5.3.2 are in an elevator cab that begins to accelerate upward, (a) Do the magnitudes of $\vec{F}_{CT}$ and $\vec{F}_{TC}$ increase, decrease, or stay the same? (b) Are those two forces still equal in magnitude and opposite in direction? (c) Do the magnitudes of $\vec{F}_{CE}$ and $\vec{F}_{EC}$ increase, decrease, or stay the same? (d) Are those two forces still equal in magnitude and opposite in direction?

Answer:

(a) they increase

(b) yes

(c) they begin to decrease slowly (the gravitational force of Earth decreases with height—negligible in an actual elevator)

(d) yes

Tension-Friction Problem#1

Statement

A block of mass $M = 3.3$ kg, connected by a cord and pulley to a hanging block of mass $m = 2.1$ kg, slides across a frictionless surface

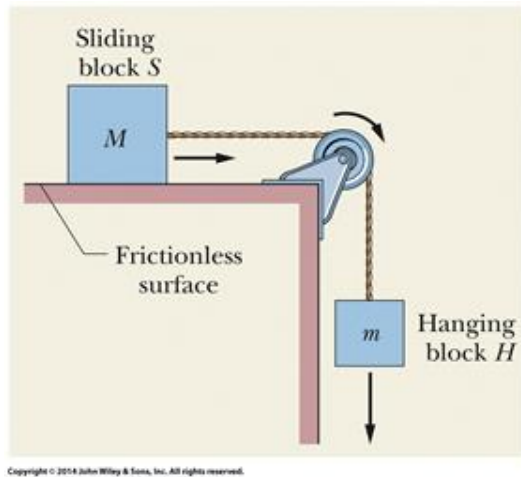


Figure 5.3.3

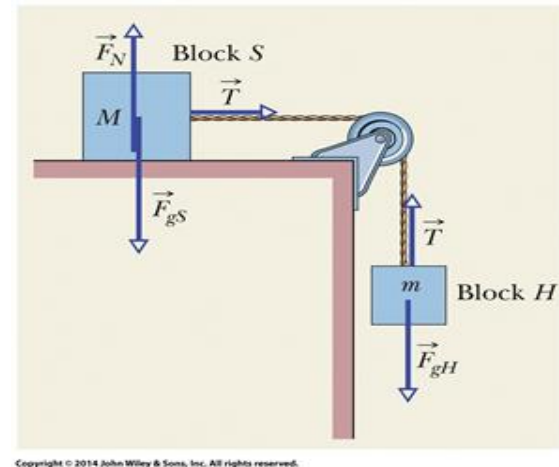
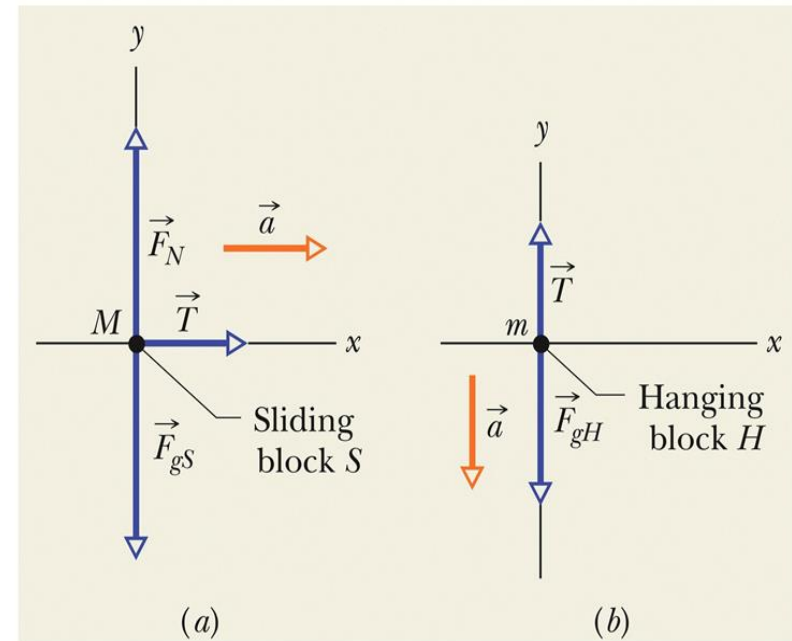


Figure 5.3.4

Tension-Friction Problem #1: Free Body Diagrams

- Draw the forces involved
- Treat the string as unstretchable, the pulley as massless and frictionless, and each block as a particle
- Draw a free-body diagram for each mass
- Apply Newton's 2nd law ($F = ma$) to each block $\rightarrow$ 2 simultaneous equations.
- Eliminate unknowns (T) that are the same, and solve for the acceleration



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Figure 5.3.5

Tension-Friction Problem #1: Force Equations

- For the sliding block:

$$T = Ma. \quad \text{Equation (5.3.4)}$$

- For the hanging block:

$$T - mg = -ma. \quad \text{Equation (5.3.6)}$$

- Combining we get:

$$a = \frac{m}{M + m} g. \quad \text{Equation (5.3.7)} \quad T = \frac{Mm}{M + m} g. \quad \text{Equation (5.3.8)}$$

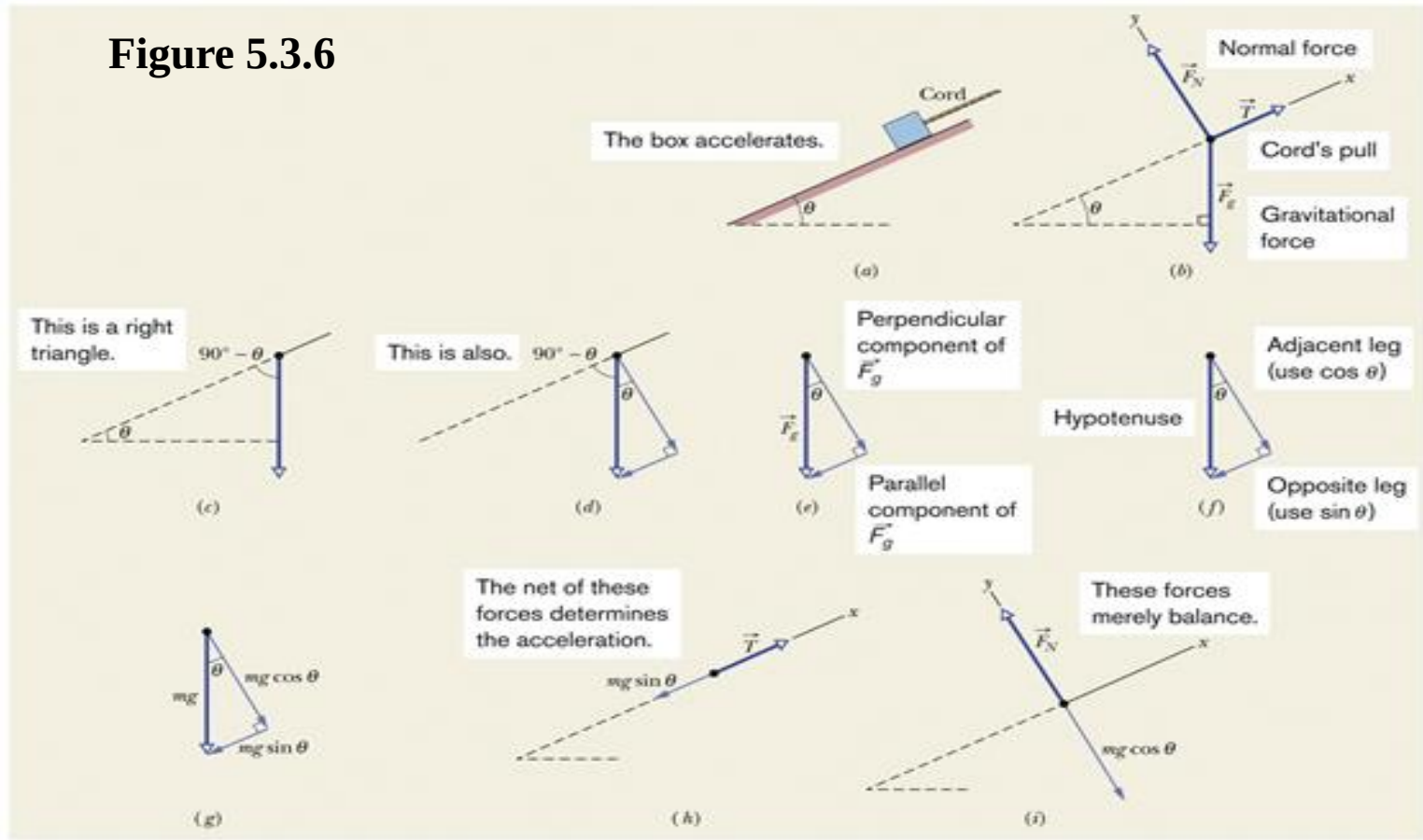
Friction-Tension Problem #1: Solution

- Plugging in we find $a = 3.8 \text{ m/s}^2$ and $T = 13 \text{ N}$
- Does this make sense? Check that dimensions are correct, check that $a < g$, check that $T < mg$ (otherwise acceleration would be upward), check limiting cases (e.g., $g = 0$, $M = 0$, $m = \infty$)

We find that our work checks out!

Friction-Tension: Block Pulled up a Ramp

Figure 5.3.6



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Chapter 5 Summary

Chap 5 Summary: Mechanics & Force

Newtonian Mechanics

- Forces are pushes or pulls
- Forces cause acceleration

Force

- Vector quantities
- $1 \text{ N} = 1 \text{ kg m/s}^2$
- Net force is the sum of all forces on a body

Chap 5 Summary: Newton's First Law & Inertial Reference Frames

Newton's First Law

- If there is no net force on a body, the body remains stationary if it is initially stationary, or continues in a straight line at a constant speed if it is in motion.

Inertial Reference Frames

- Frames in which Newtonian mechanics holds

Chap 5 Summary: Mass and Acceleration

Mass

- The characteristic that relates the body's acceleration to the net force applied to the body
- Scalar quantity

Newton's Second Law:

The net vector force on a body is equal to the body's mass times its acceleration.

- Free-body diagram represents the forces on one object

Chap 5 Summary: Particular Forces

Some Particular Forces

- Weight: mass times the acceleration due to gravity
- Normal force from a surface
- Friction along a surface
- Tension in a cord

Chap 5 Summary: Newton's Third Law

Newton's Third Law

- Law of force-pairs
- If there is a force **by B on C**, then there is a force **by C on B**:

$$F_{BC} = F_{CB} \quad \vec{F}_{BC} = -\vec{F}_{CB} \quad \text{Equation (5.3.1)}$$

Biomedical Applications: Problem 47

The Zacchini family was renowned for their human-cannonball act in which a family member was shot from a cannon using either elastic bands or compressed air. In one version of the act, Emanuel Zacchini was shot over three Ferris wheels to land in a net at the same height as the open end of the cannon and at a range of 69 m. He was propelled inside the barrel for 5.2 m and launched at an angle of 53-degrees. If his mass was 85 kg and he underwent constant acceleration inside the barrel, what was the magnitude of the force propelling him? (*Hint: treat the launch as though it were along a ramp at 53-degrees . Neglect air drag.*)

Answer: $6.4 \times 10^3 \text{N}$

Biomedical Applications: Problem 62

In shot putting, many athletes elect to launch the shot at an angle that is smaller than the theoretical one (about 42-degrees) at which the distance of a projected ball at the same speed and height is greatest. One reason has to do with the speed the athlete can give the shot during the acceleration phase of the throw. Assume that a 7.260 kg shot is accelerated along a straight path of length 1.650 m by a constant applied force of magnitude 380.0 N, starting with an initial speed of 2.500 m/s (due to the athlete's preliminary motion). What is the shot's speed at the end of the acceleration phase if the angle between the path and the horizontal is (a) 30.00-degrees, and (b) 42.00-degrees ? (Hint: Treat the motion as though it were along a ramp at the given angle.) (c) By what percent is the launch speed decreased if the athlete increases the angle from 30.00-degrees to 42.00-degrees ?

Answers: (a) 12.76 m/s, (b) 12.54 m/s, (c) 1.69%

Biomedical Applications: Problem 84

If a person drinks a can of Diet Coke before entering the doctor's office to be weighed, how much will the drink increase the weight measurement? Answer in pounds. The can has 12 US fluid ounces and the drink is flavored water with a density of 997 kilograms per cubic meter.

Answer: 0.779 lb